ST422 Report: Central England Seasonal Temperature Change

Team 3: Yueran Wang 2201370

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1 Executive summary

1.1 The client question

The brief asked for an evidence-based view of how Central England temperatures have changed across the four seasons, and what those changes mean for internal planning. Three practical questions drove the work, covering whether recent decades are measurably warmer than an earlier generation, which seasons have changed the most, and what monitoring thresholds to use in seasonal briefings. This report addresses all three.

1.2 Headline findings

- Spring temperatures averaged roughly 1.1 degrees Celsius higher in the 1991 to 2020 period than in 1961 to 1990, the largest shift of the four seasons. Summer and autumn also show clear increases. Winter points in the same direction with a smaller and less certain shift.
- Over the full historical record from 1879 to 2024, every season has been warming at a sustained long-run rate. All four seasons fall in the range of about 0.09 to 0.13 degrees Celsius per decade, with autumn warming fastest and summer warming slowest.
- The pace of warming strengthened after around 1975. Warming rates estimated over the most recent decades are consistently higher than those calculated over the full historical record in every season.
- A seasonal monitoring framework has been defined, assigning any incoming season a status of Reference, Watch, Action, or Escalation based on how far that season sits from the 1991 to 2020 reference distribution.

1.3 Recommendation

Update all internal planning benchmarks to the 1991 to 2020 reference window. Assumptions built on the 1961 to 1990 baseline are now understating typical seasonal temperatures by roughly 0.6 to 1.1 degrees Celsius depending on the season, and this gap is likely to widen as the long-run warming trend continues. The main adjustment required is a revision of existing plans that use the older baseline, since any thresholds or targets set against 1961 to 1990 conditions will be exceeded more frequently than originally intended. Implement the four-level threshold framework from the next seasonal briefing cycle to enable consistent monitoring going forward.

1.4 Evidence anchors

See Figure 1 in Section 3.2, Figure 2 in Section 4.1, Figure 3 in Section 4.2, Figure 4 in Section 4.3, and Table 1 in Section 4.4.

1.5 Key uncertainties

The winter warming signal points in the same direction as the other three seasons, but its magnitude is less certain. The range of plausible values for the winter baseline shift includes zero, so winter conclusions should be treated with more caution than spring, summer, and autumn. All findings are specific to the Central England regional aggregate and do not describe variation at a finer local scale. The monitoring thresholds are designed for practical use in seasonal briefings, and different threshold settings would trigger alerts at different rates, which is worth bearing in mind if the framework is reviewed in future.

2 Client brief and scope

2.1 What the client asked

The brief requested a clear, evidence-based assessment of seasonal temperature change in Central England using the Met Office HadCET records. The practical goal was to support three linked planning decisions, covering whether a sustained shift in seasonal temperatures has taken place over recent decades, which seasons have changed the most and whether the change looks more like a gradual trend or a distinct step shift, and what metrics and thresholds to use for internal monitoring and communication.

2.2 What this analysis covers

The analysis uses HadCET seasonal temperature records from 1879 to 2024, covering mean, minimum, and maximum temperatures across winter, spring, summer, and autumn. The primary baseline comparison is between a recent 30-year

reference period (1991 to 2020) and an earlier historical period (1961 to 1990). Long-run warming rates are estimated over the full historical window. A shorter 10-year window covering 2015 to 2024 is also used as a near-term signal to detect whether the most recent decade sits above the 30-year reference distribution.

2.3 What falls outside this scope

This report does not analyse sub-regional variation within Central England. It works with seasonal-level temperature summaries, so short-duration events such as individual heatwaves or cold spells at a daily or weekly scale are not captured. It does not make projections about future temperature levels.

3 Data and context

3.1 Where the data comes from

The analysis draws on three official seasonal temperature series from the Met Office HadCET archive (v2.1.0.0), covering seasonal mean, minimum, and maximum temperatures for the Central England region. Each series provides one value per season per year across a very long historical span. The mean series extends back to 1659, whilst the minimum and maximum series each begin in 1878. Because comparing all three temperature types requires overlapping records, all quantitative analysis in this report uses the shared period from 1879 to 2024.

3.2 What the data shows

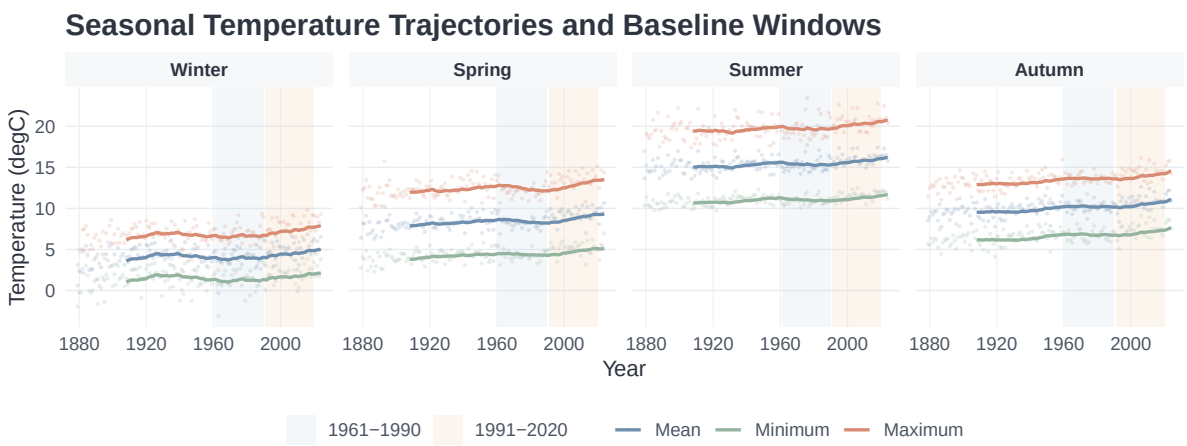

Figure 1: Annual HadCET seasonal temperatures by season, with rolling 30-year means.

Figure 1 shows four panels, one for each season. In each panel, the faint dots represent the raw temperature recorded in each individual year, and the coloured lines trace the 30-year running average for mean, minimum, and maximum temperature. The two shaded bands mark the older reference window (1961 to 1990) and the more recent one (1991 to 2020). In every panel, the running average lines rise clearly from left to right, with the upward movement becoming steeper from the 1970s onward, and both shaded windows sit visibly above the temperature levels in the early part of the record. This consistent pattern across all four seasons establishes that the changes described in Section 4 are part of a broad, sustained process and are not the product of one or two unusual years.

3.3 A note on the data

Because each data point summarises a full three-month season, shorter events within that period are not separately visible. A particularly hot fortnight in August, for instance, would contribute to the summer average but would not appear as an individual observation. A small number of historically exceptional seasons are present in the data, including winter 1963 and summer 1976. These were retained in the analysis, and later checks confirmed that they do not drive any of the headline results, so there is no need to treat them as special cases in the monitoring framework.

4 Findings

4.1 Recent decades are warmer than a generation ago

Every season and every temperature type recorded a higher average in the 1991 to 2020 period than in 1961 to 1990. Spring shows the largest increase, with its mean temperature rising by roughly 1.1 degrees Celsius. Summer and autumn also show clear upward shifts across mean, minimum, and maximum temperature. Winter is positive across all three temperature types, whilst its magnitude is harder to pin down with precision.

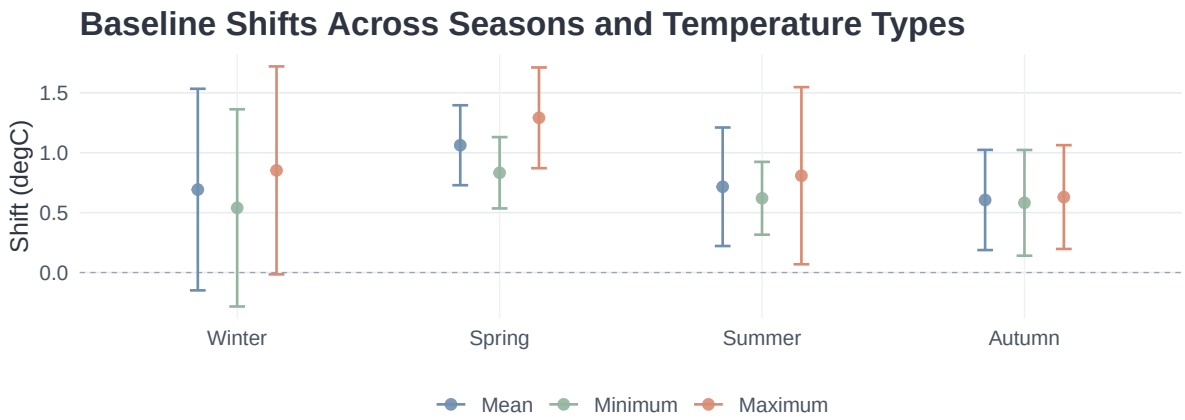

Figure 2: GLS AR(1) baseline shift estimates with 95% intervals by season and temperature type.

Figure 2 has four rows, one for each season, with three dots per row showing the estimated temperature increase for mean, minimum, and maximum. The horizontal axis shows the size of that increase in degrees Celsius, and the horizontal bars through each dot show the range of plausible values around the estimate. Every dot sits to the right of zero, confirming that all twelve season-temperature combinations were warmer in 1991 to 2020 than in 1961 to 1990. Spring's dots sit furthest right, and the bars for spring, summer, and autumn lie entirely in positive territory, whilst for winter the bars extend back close to zero, which is why the winter finding carries more caution than the other three seasons.

4.2 Warming has been sustained across the whole historical record

The full record from 1879 to 2024 shows that warming has been a persistent feature in all four seasons, not a recent development. Estimated long-run warming rates range from about 0.09 degrees Celsius per decade for summer to about 0.13 degrees per decade for autumn. Accumulated over 145 years, those rates correspond to total seasonal warming of roughly 1.3 to 1.9 degrees Celsius. All twelve combinations of season and temperature type give a positive long-run rate, with no exceptions.

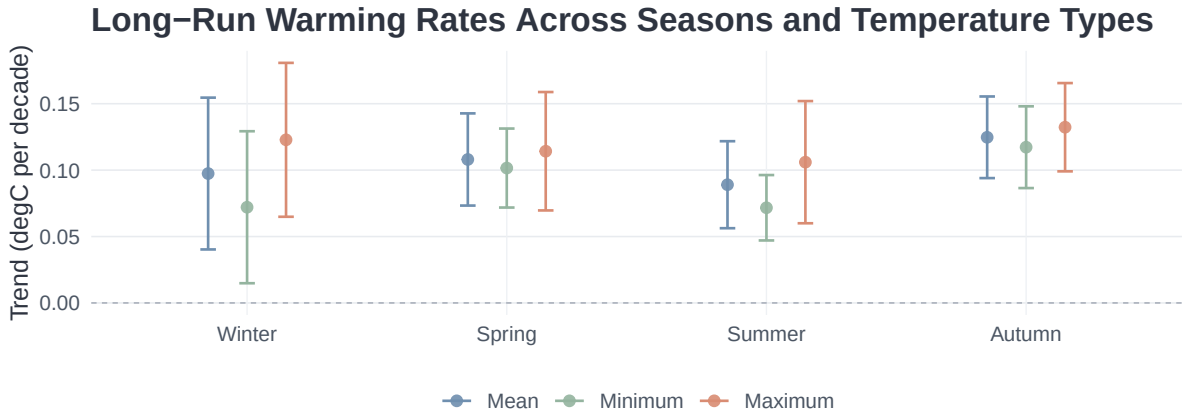

Figure 3: GLS AR(1) long-run warming rates with 95% intervals by season and temperature type.

Figure 3 follows the same layout as Figure 2, but the horizontal axis now shows the average rate of warming per decade over the full 1879 to 2024 period. The key difference from Figure 2 is that every bar in Figure 3 sits entirely to the right of zero, including winter, which in Figure 2 was the one uncertain case. Autumn tends to sit furthest right and summer furthest left, reflecting the seasonal ranking in the long-run rates. The clear separation of all bars from zero confirms that the warming pattern is a genuine long-run feature and is not simply a consequence of the particular 30-year windows chosen for comparison.

4.3 The pace of warming strengthened from the mid-1970s

The rate of warming has not been constant across the historical record. Warming rates moved into clearly positive territory from around 1975 onward across all four seasons and have remained there. Around 2010, rates in several seasons eased back slightly from their peak, but stayed positive. This acceleration into the post-1975 period is why the 1991 to 2020 reference window sits at a noticeably higher level than what preceded it.

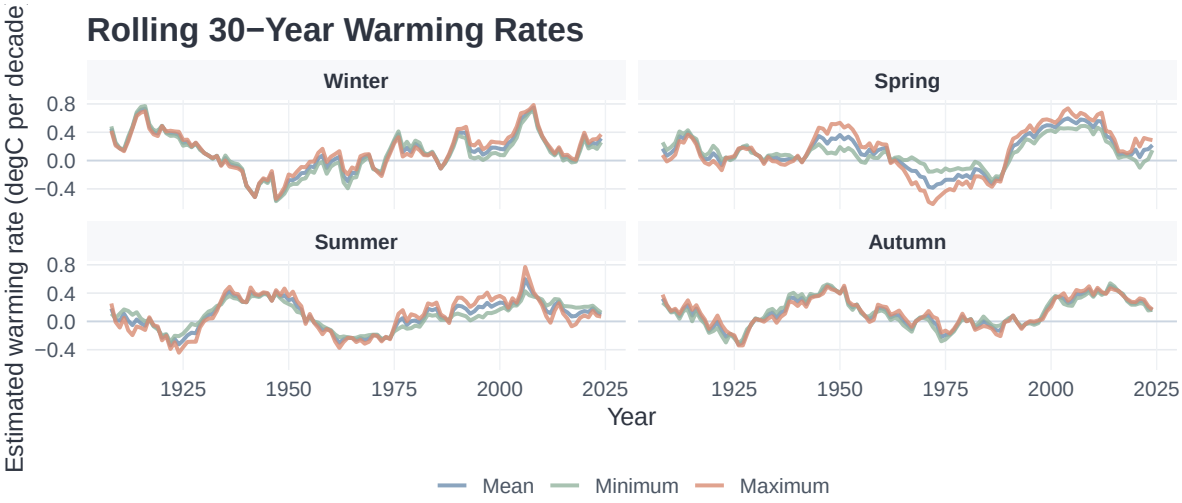

Figure 4: Rolling 30-year warming rates by season.

Figure 4 shows one panel per season, with time on the horizontal axis and the estimated warming rate within each trailing 30-year period on the vertical axis. Positive values mean that period was warming, negative values reflect a temporary cooling phase within the window. The lines for all three temperature types cross into positive territory and stay there from around 1975 to 1990 onward. Winter shows the deepest dip into negative rates around 1950, which explains why its long-run average in Figure 3 is lower than autumn's despite both finishing clearly positive by the present. The slight easing of rates after about 2010 is why the 10-year reference window (2015 to 2024) is useful as a leading check on whether current seasons are still tracking at, or above, the primary 30-year reference level.

4.4 A monitoring framework is ready to use

A practical monitoring framework has been built from the 1991 to 2020 reference distribution. For each of the twelve combinations of season and temperature type, three threshold values are defined from that distribution. P10 marks the lower boundary of the typical range, P50 is the midpoint used as the central reference for briefings, and P90 marks the upper boundary. Table 1 reports all of these for both the 30-year and the 10-year (2015 to 2024) windows side by side, making it straightforward to see whether the most recent decade already sits above the primary reference level.

Table 1: Seasonal monitoring thresholds from the 30-year window (1991 to 2020) and the 10-year window (2015 to 2024) (degC).

Season	Metric	P10 (30y)	P50 (30y)	P90 (30y)	P10 (10y)	P50 (10y)	P90 (10y)
Winter	Mean	3.12	4.91	6.19	4.50	5.69	6.56
Winter	Minimum	0.59	2.10	3.30	1.67	2.62	3.80
Winter	Maximum	5.98	7.84	9.13	7.31	8.65	9.33
Spring	Mean	8.66	9.34	10.18	8.68	9.49	10.43
Spring	Minimum	4.38	5.22	5.91	4.44	5.23	6.25
Spring	Maximum	12.47	13.50	14.57	12.91	13.74	14.70
Summer	Mean	15.32	16.01	17.28	15.75	16.42	17.31
Summer	Minimum	10.81	11.60	12.07	11.12	12.01	12.15
Summer	Maximum	19.30	20.57	22.19	20.37	20.78	22.53
Autumn	Mean	9.81	10.84	11.55	10.73	10.94	12.16
Autumn	Minimum	6.39	7.36	8.26	7.12	7.50	8.56
Autumn	Maximum	13.20	14.29	15.09	14.16	14.54	15.70

Any incoming season can immediately be assigned one of four status levels by comparing its observed value against Table 1. A season within P10 to P90 is **Reference** status, indicating conditions that fall within the established range of the 1991 to 2020 baseline. A single season outside P10 or P90 is **Watch** status, flagged for tracking in the next briefing. Two consecutive seasons outside the reference range, or a particularly large single-season breach, triggers **Action** status, calling for escalation with a seasonal impact note. If at least two of the three temperature metrics reach Action status in the same season, the framework triggers **Escalation**, prompting a cross-team planning discussion.

5 Uncertainty and limitations

5.1 Where the evidence is less certain

Winter is the season where the analysis provides the least precise picture of the recent baseline shift. Its long-run trend is clearly positive, but for the two-window comparison the range of plausible values extends close to zero, so the magnitude of winter's recent shift carries more uncertainty than spring, summer, or autumn. The monitoring thresholds in Table 1 are a practical choice, not a fixed scientific optimum. P10 and P90 were selected because they flag genuinely uncommon events without generating alerts so frequently that the framework loses its signal value, and different threshold settings would produce different alert frequencies.

5.2 What this analysis does not capture

The seasonal data does not show what happens within a season. A summer that sits within the P10 to P90 range on the seasonal average could still contain individual weeks of conditions that fall well outside normal experience. All conclusions describe conditions at the Central England regional scale and do not characterise temperature change at any specific site within the region.

5.3 What would change these conclusions

The direction of warming in all four seasons is well supported and would not be reversed by any plausible adjustment to the analysis. The main risk is on precision, particularly for winter. The statistical approach here assumes a relatively stable error structure across the full record, whilst Figure 4 shows that warming rates have themselves been changing over time. If the true error structure is more complex, some stated uncertainty ranges, especially for winter, may be slightly narrower than they should be. This does not affect the case for updating planning benchmarks, but it means winter-specific decisions should carry a small additional margin of caution.

6 Recommendations and next steps

6.1 Immediate actions

Updating all internal planning benchmarks to the 1991 to 2020 reference window is the most immediate priority. Any seasonal target or assumption currently expressed in terms of the 1961 to 1990 climate is benchmarked against conditions that were roughly 0.6 to 1.1 degrees Celsius cooler depending on the season, and this gap will widen as the long-run trend continues. Adopting the four-level threshold framework in Table 1 for the next seasonal briefing cycle requires no further analysis, since the thresholds are already computed and the status levels are already defined.

6.2 Priorities by season

Spring requires the earliest attention because its recent mean shift of roughly 1.1 degrees Celsius is the largest of the four seasons, meaning plans calibrated against the old baseline are furthest from current spring conditions. Autumn has the fastest long-run warming rate at around 0.13 degrees Celsius per decade, making it the season most likely to continue drifting upward over a five to ten year planning horizon. Winter should remain under monitoring but does not yet call for urgent revision of planning assumptions, given that the short-term evidence for its recent shift is the least certain of the four seasons.

6.3 Next steps

The first operational use of the monitoring framework should compare the most recently completed season against the thresholds in Table 1 and assign a status level. A useful secondary check at that stage is to compare the 10-year P50 values against the 30-year P50 values in Table 1. Where the 10-year midpoint already sits above the 30-year one, the organisation is routinely experiencing conditions in the upper half of the reference distribution, which reinforces the case for planning revision. If closer monitoring of within-season conditions becomes necessary, extending the analysis to monthly resolution would be the natural next step.

7 Technical Abstract

This report addresses the client question of whether Central England seasonal temperatures have shifted in recent decades, which seasons show the strongest change, and which monitoring thresholds should guide internal planning. Using Met Office HadCET seasonal mean, minimum, and maximum series (v2.1.0.0), the analysis combines EDA with season-metric specific GLS models under AR(1) errors to estimate baseline shifts (1991 to 2020 vs 1961 to 1990) and long-run trends (1879 to 2024). The headline conclusion is consistent across methods. Recent shifts are positive across all season-metric combinations, spring shows the largest baseline increases, winter is the least precise baseline case, and long-run trends are positive in all 12 combinations with post-1975 acceleration evident in sensitivity checks. The main limitations are simplified dependence modelling, visible winter-tail residual deviation, heuristic monitoring thresholds, and aggregation to seasonal regional series. These were handled by comparing OLS, AR(1), and AR(2) specifications, checking residual diagnostics directly, running segmented-trend and influential-year robustness checks, and presenting threshold rules as operational screening triggers rather than fixed optimal policy cut-offs.

8 Problem Definition and Decision Framework

8.1 Client Decision Context, Scope and Estimands

The client requires decision-ready evidence on three linked questions, whether recent seasonal temperatures have shifted relative to earlier baselines, which seasons show the strongest changes and in what form, and which monitoring metrics and thresholds should guide internal planning. The analysis targets the HadCET Central England seasonal series (Midlands representation), with year as the unit of analysis and seasonal mean, minimum, and maximum temperature (degC) as primary outcomes. For cross-metric comparability, the core window is 1879 to 2024. Two primary estimands are used, seasonal baseline shift between 1961 to 1990 and 1991 to 2020, and seasonal long-run trend in degC per decade over 1879 to 2024. Decision support is judged by coherence in direction, seasonal ranking, and uncertainty across descriptive and model-based evidence. Monitoring thresholds are operationalised in Section 13 using season-metric reference distributions.

9 Data and Pre-processing

9.1 Data Provenance, Structure, and Analytical Scope

This analysis uses three archived HadCET seasonal totals files in this repository, corresponding to Met Office (2025), *Seasonal Mean, Minimum and Maximum Central England Temperature (HadCET) series* v2.1.0.0, NERC EDS Centre for Environmental Data Analysis, doi:10.5285/ca43505702fa4eeeba4b65f1ce2c1e6a. Each file provides year-indexed seasonal temperatures (winter, spring, summer, autumn, annual) in degC. In cleaned outputs, mean covers 1659 to 2024 (366 rows), whilst minimum and maximum each cover 1878 to 2024 (147 rows).

9.2 Inclusion/exclusion Rules and Known Data Limitations

Included data are the three HadCET seasonal totals series (mean, minimum, maximum). Seasonal ranked files are excluded because they are not the underlying time-series inputs for modelling. No external joins, covariates, or spatial merges are used. Known limitations are structural. The series is a Central England regional aggregate and does not resolve sub-regional Midlands variation. Seasonal values aggregate three-month periods, so within-season short-duration extremes are not directly represented. Coverage differs by metric (mean from 1659, minimum and maximum from 1878), and the first winter entry in each series is structurally missing because previous-December data are unavailable for the first recorded year.

9.3 Pre-processing and Feature Construction

Pre-processing removed non-data header and footer lines, retained the six analytical fields `year`, `winter`, `spring`, `summer`, `autumn`, and `annual`, converted `year` to integer, converted seasonal fields to numeric, and exported cleaned series for modelling. Missingness is structural rather than random. The only missing values are the first winter entry in each series, caused by the winter definition requiring previous-December data. No imputation was applied. Potentially influential observations were flagged using two rules within each season-metric series. The first rule used robust MAD z-scores with `|z| > 3.5`. The second rule used IQR fences with lower bound `Q1 - 1.5*IQR` and upper bound `Q3 + 1.5*IQR`. This flagged 30 observations, including winter 1963, summer 1976, and autumn 2006, 2022, and 2023. Flagged values were retained and evaluated through sensitivity checks in Section 12.2. After cleaning, the mean series

contains 366 yearly records from 1659 to 2024, and the minimum and maximum series each contain 147 yearly records from 1878 to 2024. Cross-metric analysis uses the overlapping 1879 to 2024 window.

10 Exploratory Analysis and Descriptive Checks

10.1 EDA Objectives

The exploratory work addressed two main questions before any formal analysis was applied. The starting point was to examine the raw annual records alongside the long-run rolling averages and the two baseline windows, so that the direction of change and the context of the comparison could be assessed together in a single view. Once the overall direction was established, the next question was whether the pace of warming had remained stable across the record or had intensified in more recent decades.

10.2 Annual Records and Long-Run Context

Figure 5 plots the raw annual temperature records from 1879 to 2024 for each season, with rolling 30-year mean lines overlaid for each metric and the two comparison windows shaded in the background.

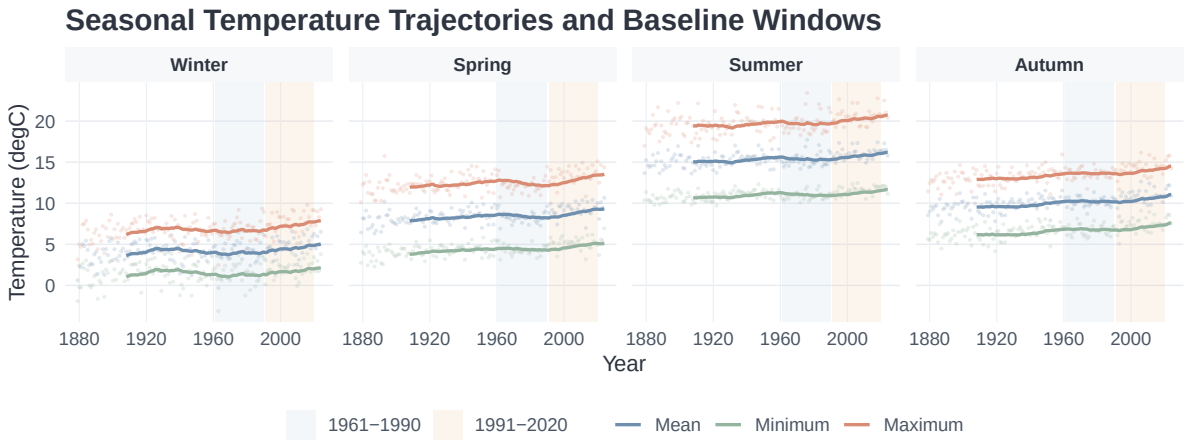

Figure 5: Annual HadCET seasonal temperatures by season, with rolling 30-year means.

The four panels correspond to the four seasons, each showing faint annual data points alongside three rolling 30-year mean lines for mean, minimum, and maximum temperature distinguished by colour, with the two baseline windows highlighted by shaded bands. The rolling means rise in every panel, with the rate of increase steepening in the later decades, and both shaded windows sit on the upward portion of each curve rather than at a stationary level. The consistent rise across all panels before, during, and after both windows establishes that the baseline shift is part of a sustained long-run process.

10.3 Rolling Warming Rates Through Time

Figure 6 below tracks the estimated warming rate within each trailing 30-year window. This turns the long-run trajectories in Figure 5 into a time-varying rate measure, so the record can be assessed not only for direction but also for whether the pace of warming has remained stable or intensified. It's clear that the pace of warming has not been constant over time. Around 1950, winter is the only season with a clearly negative rolling warming rate, indicating a temporary cooling phase within the 30-year window, whereas spring, summer, and autumn are already warming and follow broadly similar paths across most of the record.

Thereafter, all four seasons move into positive territory after about 1975, and negative rates become rare after 2000. At the same time, the three temperature metrics do not move equally closely together across seasons. Autumn shows the greatest overlap between mean, minimum, and maximum temperature, whilst spring shows the least. Winter and summer are more tightly grouped before about 1950, but separate more clearly afterwards. Therefore, these patterns indicate that temperatures have not only risen in recent decades, but that the rate of increase has also strengthened over time.

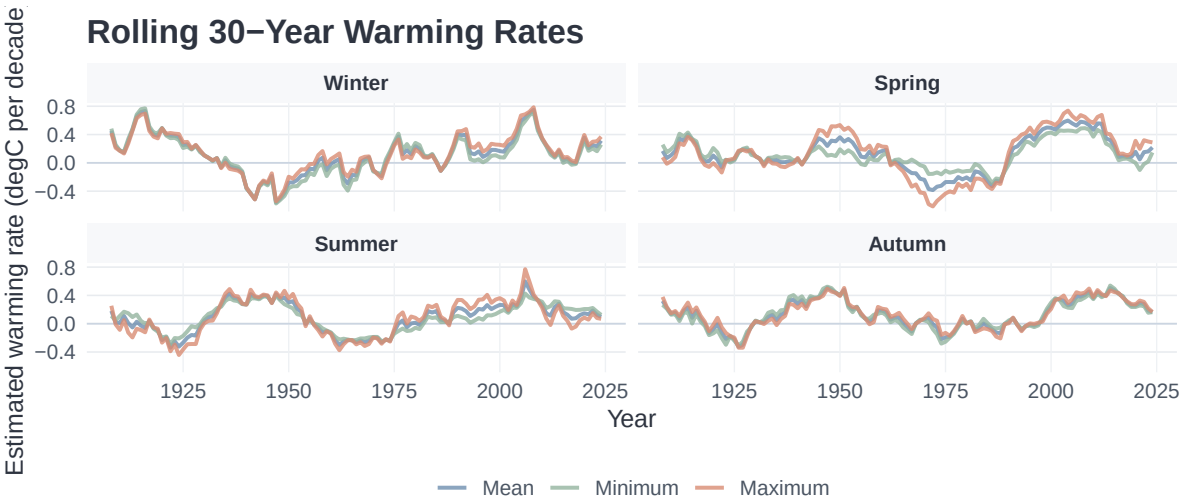

Figure 6: Rolling 30-year warming rates by season.

11 Methodology of Modelling

11.1 Model choice & Assumptions

The main models are fitted using GLS with an AR(1) error structure. This choice allows for year-to-year dependence in seasonal temperature series while remaining simple, interpretable, and directly comparable across seasons and temperature metrics.

11.2 GLS AR(1): Baseline Shift Model

The shift model estimates **how much warmer the 1991 to 2020 period was compared with 1961 to 1990 for a given season and temperature type**. For each of the 12 season-metric combinations, the model is fitted separately to 60 observations, comprising 30 annual values from the historical window and 30 from the recent window. The formal specification is

$$\text{temp}_t = \beta_0 + \beta_1 \cdot \mathbf{1}[\text{recent}_t] + \varepsilon_t, \qquad \varepsilon_t = \rho\,\varepsilon_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma^2)$$

Each symbol has a direct interpretation:

- temp_t is the observed seasonal temperature in year t, the raw number in the dataset.
- β_0 is the mean temperature in the historical window, the **baseline level** against which the recent period is compared.
- The indicator $\mathbf{1}[\text{recent}_t]$ acts as a switch, equalling one for years 1991 to 2020 and zero for years 1961 to 1990.
- β_1 is therefore the primary estimand, representing **how much higher the recent window mean is compared with the historical window mean**, in degrees Celsius.
- The residual ε_t is the difference between that year's observed temperature and the average level for its window, and is modelled as an AR(1) process rather than assumed to be independent across years.
- The parameter ρ measures how much this year's "above or below average" carries over into next year. If ρ is positive, a year that is warmer than its window average makes the next year more likely to be warmer than its window average too, so adjacent observations are dependent rather than independent.
- η_t is the white-noise innovation remaining after removing that dependence, assumed to follow a normal distribution with mean zero and variance σ^2.

The four unknowns β_0, β_1, ρ, and σ^2 are estimated jointly using REML in R. The model is fitted 12 times in total, once for each combination of four seasons and three temperature types (mean, minimum, and maximum). Each fit uses an entirely separate set of 60 observations and produces its own estimates of β_1, ρ, and the associated confidence interval. Table 2 and Figure 7 collect all 12 results.

Table 2: GLS AR(1) baseline shift estimates (degC) with 95% CIs.

Season	β_1: Mean	β_1: Min	β_1: Max
Winter	+0.692 [-0.149, +1.534]	+0.540 [-0.283, +1.363]	+0.853 [-0.015, +1.721]
Spring	+1.063 [+0.729, +1.396]	+0.833 [+0.535, +1.130]	+1.292 [+0.871, +1.712]
Summer	+0.716 [+0.221, +1.211]	+0.620 [+0.316, +0.924]	+0.808 [+0.069, +1.548]
Autumn	+0.606 [+0.188, +1.024]	+0.582 [+0.141, +1.023]	+0.630 [+0.197, +1.063]

Summer minimum in Table 2 is used here as a worked example to illustrate how the model works:

$$\text{temp}_t = 10.92 + 0.62 \cdot \mathbf{1}[\text{recent}_t] + \varepsilon_t, \qquad \hat{\rho} = 0.195$$

The historical-window mean is 10.92degC and the estimated shift is 0.62degC, which means that the average summer minimum temperature in 1991 to 2020 is about 0.62 degC higher than in 1961 to 1990. The estimated $\hat{\rho}$ of 0.195 indicates mild year-to-year dependence in the residual deviations, so adjacent summers are somewhat related rather than fully independent. The 95% confidence interval [0.316, 0.924] lies entirely above zero, which means that the recent-window increase remains clearly positive even after the AR(1) correction has widened the uncertainty interval.

Table 2 and Figure 7 point in the same direction across almost all season-metric combinations. Every point estimate is positive, which indicates that the 1991 to 2020 window is warmer than 1961 to 1990 for 3 temperature types in every season. The clearest evidence appears in spring, where all three confidence intervals lie comfortably above zero and the estimated shifts are also the largest in magnitude. Summer and autumn show the same positive pattern, although with slightly smaller shifts than spring. Winter is the least certain case. Its point estimates are still positive throughout, but the lower bounds of all three intervals fall just below zero, so the evidence for a winter baseline shift is relatively weaker than other seasons.

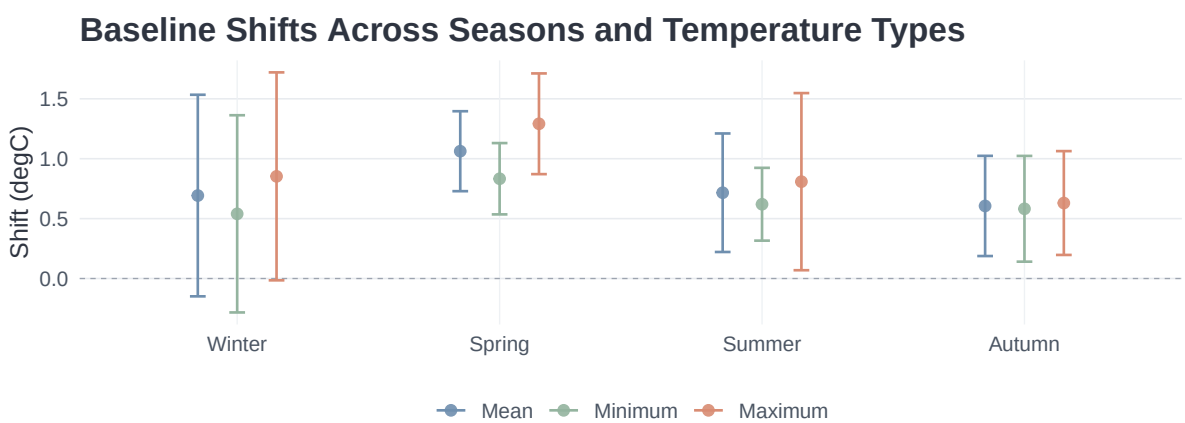

Figure 7: GLS AR(1) baseline shift estimates with 95% intervals by season and temperature type.

11.3 GLS AR(1): Long-Run Trend Model

The long-run trend model asks a different question from the baseline shift model. Instead of comparing two fixed periods, it asks how much temperature tends to change when year increases by one.

$$\text{temp}_t = \beta_0 + \beta_1 \cdot (t - \bar{t}) + \varepsilon_t, \qquad \varepsilon_t = \rho\,\varepsilon_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma^2)$$

Here, β_0 is the fitted temperature level at the average year of the series. β_1 is the annual slope, so $\beta_1 \times 10$ tells us how much temperature is estimated to increase per decade. The term $(t - \bar{t})$ means that the model measures each year relative to the middle of the observed time period, because this leaves the fitted slope unchanged, makes the intercept easier to interpret, and reduces collinearity between the intercept and slope. The definitions of ε_t, ρ, η_t, and σ^2 are otherwise unchanged from the shift model. Each season-metric combination is again fitted separately, now using the full 1879 to 2024 record, so the estimated slope is less sensitive to the exact placement of the two baseline windows and makes use of substantially more historical information. Spring maximum in Table 3 is used here as a worked example:

$$\text{temp}_t = 12.55 + 0.0114 \cdot (t - 1951.5) + \varepsilon_t, \qquad \hat{\rho} = 0.141$$

Table 3: GLS AR(1) long-run warming rates (degC per decade).

Season	$\beta_1 \times 10$: Mean	$\beta_1 \times 10$: Min	$\beta_1 \times 10$: Max
Winter	+0.097 [+0.040, +0.155]	+0.072 [+0.015, +0.129]	+0.123 [+0.065, +0.181]
Spring	+0.108 [+0.073, +0.143]	+0.102 [+0.072, +0.131]	+0.114 [+0.070, +0.159]
Summer	+0.089 [+0.056, +0.122]	+0.072 [+0.047, +0.096]	+0.106 [+0.060, +0.152]
Autumn	+0.125 [+0.094, +0.155]	+0.117 [+0.086, +0.148]	+0.132 [+0.099, +0.166]

$\beta_0 = 12.55$ is the fitted spring maximum temperature at the average year of the record, and $\beta_1 = 0.0114$ means that the fitted spring maximum temperature rises by about 0.0114 degrees Celsius when year increases by one. Multiplying that annual slope by ten gives 0.114 degrees Celsius per decade, which is the value reported in Table 3. The estimated $\hat{\rho}$ of 0.141 is clearly positive, so year-to-year dependence is not negligible in this series. The 95% confidence interval [0.07, 0.159] still remains entirely above zero, so the long-run warming signal remains positive after the AR(1) correction.

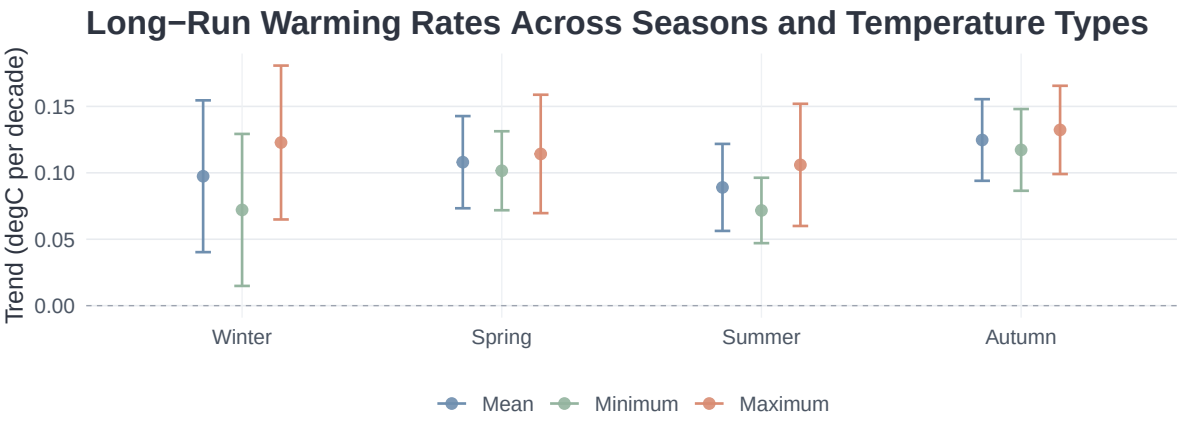

Figure 8: GLS AR(1) long-run warming rates with 95% intervals by season and temperature type.

Figure 8 confirms positive long-run warming across all four seasons and all three temperature metrics, with all 12 confidence intervals lying above zero. Autumn records the fastest mean warming rate at +0.125 degC per decade (95% CI $[+0.094, +0.155]$), while summer is the slowest at +0.089 ($[+0.056, +0.122]$). Worth highlighting is the winter result. Over the 60-year comparison window, the baseline shift was relatively weak because the confidence interval ($[-0.149, +1.534]$) straddles zero. Extending the analysis to the full 1879 to 2024 record resolves this, giving a clear positive trend of +0.097 degC per decade ($[+0.040, +0.155]$), which is a meaningful finding for a season that looked ambiguous in the earlier model.

Figure 6 helps explain why the long-run warming rates differ across seasons. A season tends to show a higher overall warming rate when its rolling 30-year temperature trend spends more time in positive territory. Autumn follows this pattern most clearly. Winter is the opposite case, with a deep and prolonged cooling phase through much of the mid-twentieth century that pulls down the long-run average and produces the weakest full-period warming rate of the four seasons. Spring and summer sit in between. This contrast also accounts for one of the more interesting findings in the results, spring records the largest recent baseline shift in Table 2 at +1.063 degC ($[+0.729, +1.396]$), yet this does not translate into the highest long-run warming rate, while autumn's smaller recent shift of +0.606 degC ($[+0.188, +1.024]$) comes with the strongest long-run average warming rate.

These intervals should be treated as uncertainty conditional on the current model. They do not account for model misspecification such as non-linearity in long-run warming, nor for structural variance change across the record, so they are best read as a floor on uncertainty rather than a complete picture.

11.4 Serial Dependence in Baseline and Long-Run Models

Figure 9 reports the estimated AR(1) coefficients, $\hat{\rho}$, for both the baseline shift model and the long-run trend model. These coefficients summarise how strongly residual deviations carry over from one year to the next after the mean structure has been fitted.

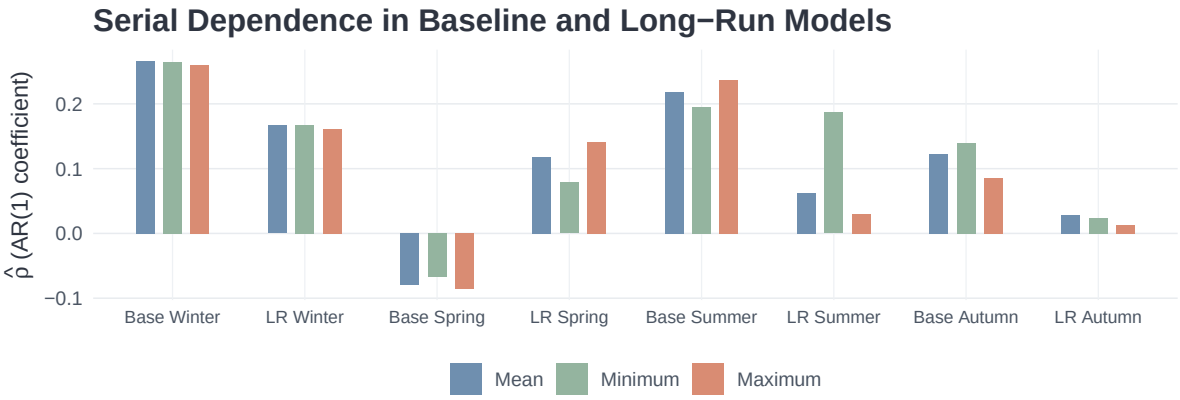

Figure 9: Estimated AR(1) serial dependence coeff from the baseline shift and LR trend models by season.

From Figure 9 we can see that serial dependence is not equally strong across the seasonal series. In the baseline shift model, winter and summer have the clearest positive AR(1) coefficients, with values mostly around 0.20 or above across the three temperature metrics, so year-to-year persistence is substantial enough to matter for inference. By contrast, baseline spring lies close to zero and is slightly negative, whilst long-run autumn is also very close to zero. Coefficients of that size should be interpreted cautiously, because they may reflect weak or unstable dependence rather than a statistically meaningful pattern. Autumn in the baseline model remains positive but is clearly weaker than winter and summer. The long-run trend model shows that dependence is still present after the trend has been fitted, especially in winter and in summer minimum, but the pattern is more mixed than in the baseline model. A notable feature is that long-run summer minimum remains clearly higher than the corresponding mean and maximum coefficients, which suggests that residual persistence is not distributed uniformly across the summer temperature range.

Overall, Figure 9 suggests that the AR(1) assumption is worth retaining as a practical working specification, because several series show clear positive persistence and ignoring it would risk understating uncertainty. At the same time, the near-zero cases show that dependence is not uniformly strong in every season, so AR(1) should be treated as an adequate overall correction rather than a perfect description of every individual series.

11.5 Model Comparison and Comparison Criteria

This section compares three dependence specifications under the same mean structure for each model family. OLS is included as the baseline specification where residuals are treated as independent, so it provides the simplest reference for fit and interval width. GLS AR(1) is included because the earlier dependence diagnostics reported non-negligible year-to-year persistence in several season-metric series. GLS AR(2) is included as a higher-order alternative to test whether a second lag materially improves fit once first-order dependence has already been modelled. For the baseline shift model, the mean structure remains the historical versus recent indicator. For the long-run trend model, the mean structure remains the centred linear year term.

The comparison records average AIC, average BIC, and average confidence-interval width for both model families. For the long-run trend family, average rolling one-step-ahead RMSE and its standardised form (NRMSE = RMSE/SD) are also reported because one-step forecasting is directly defined for a continuous time index and scale-normalised error supports cross-series comparison. Table 4 reports the baseline-shift family and Table 5 reports the long-run trend family.

Table 4: Model comparison: baseline-shift specification.

Model	Avg AIC	Avg BIC	Avg CI for β_1
OLS	156.562	162.845	0.913
AR(1)	157.095	165.472	1.022
AR(2)	158.186	168.657	0.927

Table 5: Model comparison: long-run trend specification.

Model	Avg AIC	Avg BIC	Avg CI for $\beta_1 \times 10$	Avg roll RMSE	Avg roll NRMSE
OLS	388.663	397.614	0.072	0.953	0.911
AR(1)	389.137	401.071	0.079	0.952	0.911
AR(2)	390.030	404.948	0.084	0.953	0.911

Since average AIC, average BIC, rolling one-step RMSE, and rolling NRMSE are all very close across OLS, AR(1), and AR(2), the metric differences are too small to identify a single dominant model. The deciding consideration is therefore the dependence evidence in Figure 9, where several series show residual persistence around 0.2, making the OLS independence assumption less credible for inference. AR(2) adds complexity but delivers little practical improvement over AR(1) on the reported metrics. For this reason, AR(1) is retained as the working specification, with OLS as a benchmark and AR(2) as a sensitivity alternative.

12 Validation and Robustness

The conclusions in Section 11.2 and Section 11.3 rely on the AR(1) model assumptions holding reasonably well and on the results not being driven by a small number of specific data points or modelling choices. This section checks both of those conditions directly, so that the reported confidence intervals and seasonal rankings can be taken at face value for decision use.

12.1 Residual Diagnostics

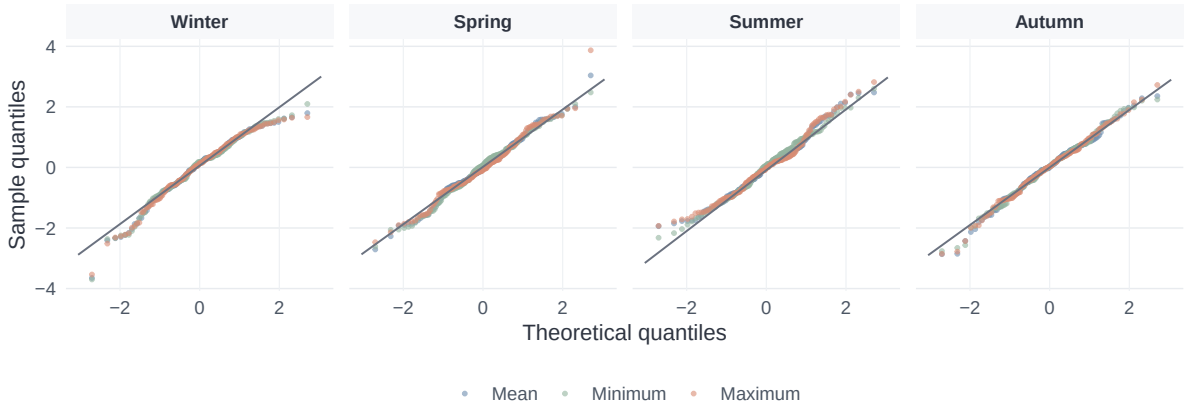

Figure 10: Q-Q plots of GLS AR(1) normalised trend residuals by season and temperature type.

Figure 10 checks whether the model residuals follow the normal distribution closely enough to trust the reported 95% confidence intervals. Across all four seasons, residuals track the reference line well through the central quantiles, which is where most of the estimation information lies, so the intervals can be taken at face value for the bulk of the distribution. The one area requiring care is the lower winter tail, consistent with the extreme winter years flagged in Section 9.3, spring and summer show mild upper-tail deviation but nothing that points to a systematic problem. In short, the normality assumption holds well enough for practical inference, with the caveat that winter tail-region estimates should be interpreted with slightly more caution.

12.2 Robustness Checks

Two robustness checks are applied to the long-run findings. The first uses a segmented AR(1) specification with a fixed 1975 breakpoint to test whether the warming conclusion holds once the model allows for a faster post-1975 slope. The second re-estimates the long-run trends after excluding the influential observations identified in Section 9.3 to confirm that the headline result is not driven by a small set of extreme years.

12.2.1 Segmented Linear Trend Sensitivity (Breakpoint at 1975)

Figure 11 compares the baseline long-run AR(1) trend against a segmented AR(1) specification with a fixed breakpoint at 1975. The breakpoint is set with direct reference to Figure 6, which shows that warming rates begin to accelerate after around 1975 across all seasons. This check tests whether allowing a faster post-1975 slope changes directional conclusions or seasonal ranking.

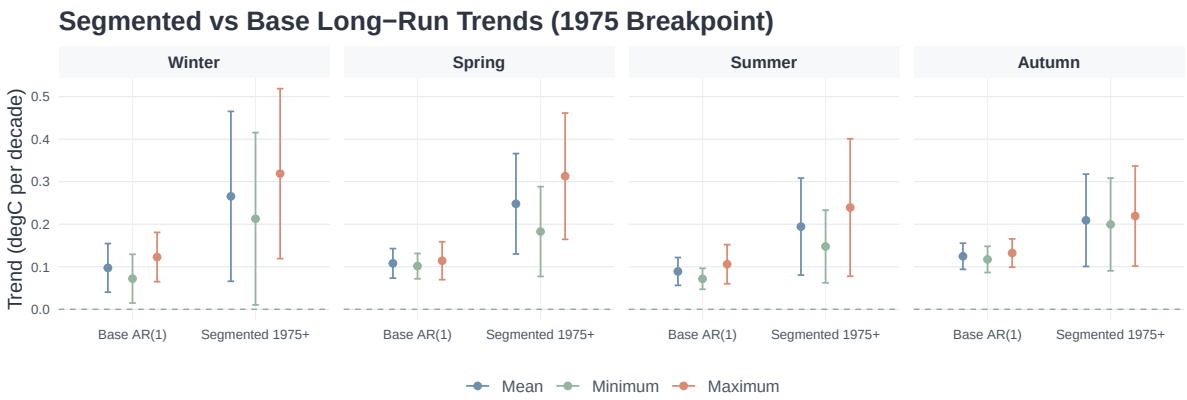

Figure 11: Segmented-trend sensitivity with a fixed 1975 breakpoint by season and temperature type.

Figure 11 shows a consistent pattern across all four seasons and all three temperature metrics, where the post-1975 segmented slopes are higher than the corresponding base AR(1) full-period slopes. The direction remains positive throughout, so the core warming conclusion is unchanged. The uplift is strongest in winter and spring, with particularly large increases in maximum temperature, whilst summer and autumn also rise in the same direction. This means the seasonal ranking of warming pace is not fixed across specifications. Full-period slopes are more balanced, whereas post-1975 slopes are more strongly led by winter and spring. Confidence intervals are wider under the segmented specification because the model estimates extra structure from a shorter effective period, yet the displayed intervals remain above zero. Therefore, the warming direction is robust and the warming rate accelerates after 1975.

12.2.2 Sensitivity to Influential Extreme Years

Figure 12 evaluates sensitivity to influential points identified in Section 9.3. The excluded observations are winter 1963, summer 1976, autumn 2006, autumn 2022, and autumn 2023.

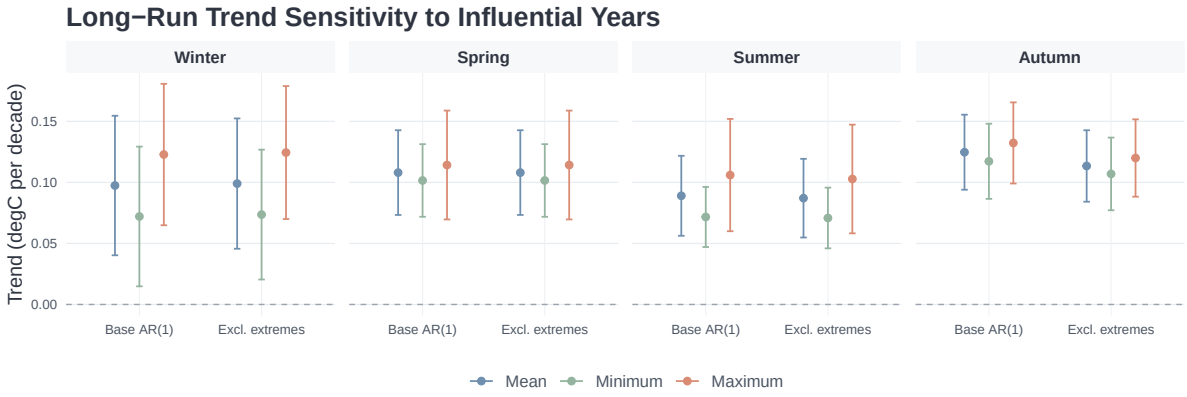

Figure 12: Sensitivity of long-run AR(1) trends to exclusion of flagged influential year-season observations.

Figure 12 indicates that excluding the flagged influential observations has only minor impact on the long-run trend estimates. Point estimates under the exclusion specification remain close to the base AR(1) estimates for every season-metric combination, and the confidence intervals overlap strongly. No directional reversal appears and the relative seasonal pattern remains stable. This indicates that the reported long-run warming trends are not driven by a small set of extreme years.

13 Monitoring and Material-Change Thresholds

Seasonal mean is used as the primary monitoring metric, and two baseline windows are used for monitoring. The 30-year window (1991–2020) is the primary decision baseline for all formal threshold calculations, anchored to the period when warming rates turned consistently positive across all four seasons (Figure 6). The 10-year window (2015–2024) serves as a short-horizon signal, rates in several seasons have moderated somewhat since around 2010. Comparing the two sets of thresholds shows directly whether the most recent decade has already shifted beyond the 30-year reference.

P10 and P90 are used as the reference band boundaries. Each marks the outer 10 per cent of the reference distribution, so a season falling below P10 or above P90 is a genuinely uncommon event relative to the 1991 to 2020 baseline, occurring about once in ten seasons under stable conditions. This frequency is low enough to carry a clear signal but common enough to remain sensitive to gradual change.

Table 6 reports P10, P50 (as reference), and P90 for each season-metric combination from the 30-year and 10-year reference windows. A consistent upward shift in the 10-year values relative to the 30-year values would confirm that the most recent decade already sits above the primary reference distribution, making the 30-year P90 threshold increasingly likely to be exceeded in coming seasons. Therefore, a three-level status classification is used for planning purposes: - **Reference**: Oservation within P10 to P90. Continue routine monitoring. - **Watch**: Single-season value outside P10 or P90. Flag in briefing and track next season. - **Action**: Two consecutive seasonal breaches beyond P10 or P90, escalate with a seasonal impact note.

Table 6: Seasonal monitoring thresholds from the 30-year window (1991 to 2020) and the 10-year window (2015 to 2024) (degC).

Season	Metric	P10 (30y)	P50 (30y)	P90 (30y)	P10 (10y)	P50 (10y)	P90 (10y)
Winter	Mean	3.12	4.91	6.19	4.50	5.69	6.56
Winter	Minimum	0.59	2.10	3.30	1.67	2.62	3.80
Winter	Maximum	5.98	7.84	9.13	7.31	8.65	9.33
Spring	Mean	8.66	9.34	10.18	8.68	9.49	10.43
Spring	Minimum	4.38	5.22	5.91	4.44	5.23	6.25
Spring	Maximum	12.47	13.50	14.57	12.91	13.74	14.70
Summer	Mean	15.32	16.01	17.28	15.75	16.42	17.31
Summer	Minimum	10.81	11.60	12.07	11.12	12.01	12.15
Summer	Maximum	19.30	20.57	22.19	20.37	20.78	22.53
Autumn	Mean	9.81	10.84	11.55	10.73	10.94	12.16
Autumn	Minimum	6.39	7.36	8.26	7.12	7.50	8.56
Autumn	Maximum	13.20	14.29	15.09	14.16	14.54	15.70

14 Limitations, Validity, and Alternatives

14.1 Threats to Validity and Limitations

Internal Validity: Section 11.4 showed that $\hat{\rho}$ was generally around 0.1 to 0.2 across the fitted series. AR(1) therefore fits the climatic intuition that adjacent years are not fully independent, but it does not follow that the effect of ρ is large enough to materially change every inference. At the same time, AR(1) captures only one-step dependence and AR(2) extends this only slightly. Neither is sufficient to guarantee that more complex, longer-range, or non-linear residual structure has been fully represented.

External Validity: The conclusions apply to the HadCET Central England seasonal series and to annual seasonal temperature conditions at that regional scale. Their generalisability is limited in two ways. First, they do not directly establish finer spatial variation within the Midlands. Second, the dataset compresses temperature information into annual seasonal values and does not retain monthly resolution, so its temporal granularity is therefore limited, which makes it more suitable for season-level analysis than for finer-grained temperature variation.

Construct Validity: The monitoring framework in Section 13 is intentionally heuristic. The use of P10 and P90 as thresholds, together with the 30-year and 10-year reference windows, was chosen for interpretability and operational usefulness in internal briefing rather than because these settings are the unique mathematical optimum. P10 and P90 are preferable to P25 and P75 for this purpose because they do not trigger as frequently, but they still represent an empirical compromise rather than a formally optimal cut-off. The same applies to the two reference windows. The

30-year window provides a longer baseline and the 10-year window provides a recent signal, yet both involve judgement rather than a single objectively correct choice.

14.2 Failure Mode That Would Change the Conclusion

The most plausible failure mode is that the model assumptions do not match the true error structure closely enough. Both Sections 11.2 and 11.3 assume that the innovation term η_t is approximately normal and use normal approximations to construct standard errors and 95% confidence intervals. If the true innovations are more skewed, more heavy-tailed, or more prone to extreme values than this assumption allows, then the interval estimates may misstate uncertainty. Section 12.1 suggests that the central residual distributions are broadly acceptable in most season-metric combinations, but the winter tail still departs visibly from the reference line. This means that the direction of warming may remain credible whilst the precision of some interval statements, especially in winter, is less secure than the model formally implies.

14.3 Alternative Approaches and Conditions for Preference

A plausible alternative is a dynamic linear model, which allows both the underlying trend and the error structure to evolve through time instead of imposing one fixed slope and one fixed dependence pattern over the full record. Relative to the GLS long run trend model in Section 11.3, this approach is better suited to the changing warming rates shown in Figure 6, and it would allow changes in the mean structure and residual dependence to be handled within one time-series framework.

It was not chosen here because the report required a common set of results that could be compared directly across seasons and temperature metrics, and the baseline-shift model together with the single long run slope provided a clearer decision-ready summary for that purpose. A dynamic linear model would become preferable if the main objective shifted from estimating recent contrasts and full-period average change to tracking how the pace of warming and the dependence structure evolve through time.

15 Traceability and Reproducibility

15.1 Claim-to-Evidence Map for Decision 1, 2, and 3

Claim 1. Recent decades are warmer than the earlier baseline in most seasons. (Table 2, Figure 7, Section 11.2)

Claim 2. Long-run seasonal warming is positive across all four seasons and across mean, minimum, and maximum temperature (Table 3, Figure 8, Section 11.3_

Claim 3. The warming pattern is sustained over the long run and is not adequately described as a single one-off shift (Figure 5, Figure 6, Figure 11, Sections 10.2, 10.3, and 12.2.1)

Recommendation 1. Internal seasonal briefing should use a threshold-based monitoring framework built from recent seasonal reference distributions (Figure 6, Figure 8, and Figure 11, Section 13)

15.2 Simplifications Planned for the Client-Facing Report

[[PENDING_CLIENT_TRANSLATION]]

15.3 Repository Structure, Run Sequence, Environment, and Determinism

[[PENDING_REPRODUCIBILITY: finalise after code rerun and dependency lock verification.]]

15.4 Output Inventory and Remaining Reproducibility Risks

[[PENDING_REPRODUCIBILITY]]

16 Appendix: Supplementary EDA Diagnostics